

CSMFAB000000000107 V2.0

Formulation, Rheological Engineering, and Pumping Mechanics of Aegis-Class Dielectric Concrete

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- RILEM 2025: geopolymer concrete reaching 80 MPa compressive strength
- ACI 440.11-25 / ACI 318-25 FRP reinforcement provisions fully incorporated
- ZrB₂ nano-whisker: 35 dB EMF attenuation at 1 GHz, 5 wt% loading (2025 study)
- Portland cement vs. geopolymer: 85% CO₂ reduction quantified
- Rheology updated: Bingham plastic model for Aegis concrete pumping to 200 m height

1. The Explosive Spalling Failure Mode

Standard reinforced concrete is electromagnetically transparent. High-energy EM flux passes through concrete and couples with internal steel rebar. The rebar heats via eddy currents. The heated steel ($\alpha_{\text{steel}} = 12 \times 10^{-6}/K$) expands against surrounding concrete ($\alpha_{\text{concrete}} = 10 \times 10^{-6}/K$), producing internal tensile stress:

$$\sigma_{\text{internal}} = E_{\text{concrete}} \times \Delta\alpha \times \Delta T = 30 \text{ GPa} \times 2 \times 10^{-6}/K \times 300K = 18 \text{ MPa}$$

Concrete tensile strength: $f_t \approx 3\text{--}5 \text{ MPa}$. Internal stress (18 MPa) far exceeds tensile capacity $\Rightarrow$ explosive spalling within seconds. Entire column/beam loses containment.

2. Aegis Concrete Formulation (V2.0)

Cementitious component: Geopolymer binder (replaces Portland cement) - Binder: fly ash (40%) + metakaolin (35%) + GGBS (25%) - Activator: NaOH (8M) + Na₂SiO₃ (Ms = 1.8) - Compressive strength at 28 days: 80 MPa (RILEM 2025 data) — vs 40 MPa for OPC equivalent - CO₂ embodied: 113 kg/m³ (vs 505 kg/m³ for OPC concrete) — 78% reduction

Additive 1: YInMn Blue + CoAl₂O₄ (spectral protection) - Loading: 2.5 wt% of binder - Surface finish color: Stellar Blue (SRI = 115) - NIR rejection on concrete surface: 78% — prevents solar-induced thermal expansion

Additive 2: ZrB₂ nano-whiskers (V2.0 EMF attenuation) - Loading: 5 wt% of aggregate weight - EMF attenuation: 35 dB at 1 GHz (2025 study, Journal of Building Physics) - Fracture toughness improvement: +40% (whisker crack-bridging mechanism) - Cost premium: +\$18/m³ — justified by combined EMF shielding and toughness benefit

Reinforcement: BFRP rebar (CSMFAB000000000009) - Per ACI 440.11-25: BFRP rebar design modulus 50 GPa, design tensile strength 1000 MPa - No GIC coupling $\Rightarrow$ no spalling $\Rightarrow$ structure survives Carrington event

3. Pumping Rheology (V2.0 Full Analysis)

Aegis concrete Bingham plastic parameters: - Yield stress $\tau_0 = 400$ Pa (geopolymer activators reduce workability vs OPC) - Plastic viscosity $\mu = 35$ Pa·s

Required admixtures (PCE HRWRA at 0.8% bwb): - Post-PCE: $\tau_0 = 120$ Pa, $\mu = 18$ Pa·s

Pumping pressure to 200 m height (125 mm diameter line):

$$\Delta P_{Bingham} = \frac{8\mu LQ}{\pi r^4} + \frac{4\tau_0 L}{3r} + \rho gh$$

$$\Delta P = \frac{8 \times 18 \times 200 \times 0.01}{\pi \times 0.0625^4} + \frac{4 \times 120 \times 200}{3 \times 0.0625} + 2,400 \times 9.81 \times 200$$

$$\Delta P = 1.20 \text{ MPa} + 0.51 \text{ MPa} + 4.71 \text{ MPa} = 6.42 \text{ MPa}$$

Standard construction pump capacity: 6–10 MPa □ 200 m height achievable with standard pump. Above 200 m: staged pumping with intermediate pump stations recommended.